GATE DA 2026
Probability and Statistics

Practice Questions | MCQ, MSQ and NAT

Questions arranged in the supplied source order

Instructions: Read each question carefully. For NAT questions, enter the requested numerical value and apply the rounding instruction stated in the question.

Q1 NAT 2 Marks

Let $A_{5\times5}$ be a matrix such that each of its elements follows Bernoulli($p = 0.50$) distribution independently.

The probability that the row-sum of the second row and the column-sum of the third column are both equal to 3 is _____. (Rounded off to two decimal places)

Q2 NAT 2 Marks

Let X be a random variable that follows Uniform$(-1, 1)$ distribution. The conditional distribution of the random variable Y given $X = x$ is the Uniform$(x^2 - 0.1, x^2 + 0.1)$ distribution.

The value of correlation(X, Y) is _____. (Answer in integer)

Q3 NAT 2 Marks

For a given data set $\{x_1, x_2, \ldots, x_n\}$, where $n = 100$, it is known that

$$\frac{1}{2000}\sum_{i=1}^{n}\sum_{j=1}^{n}(x_i - x_j)^2 = 99$$

Let us denote $\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$

The value of $\frac{1}{99}\sum_{i=1}^{n}(x_i - \bar{x})^2$ is _____. (Answer in integer)

Q4 NAT 2 Marks

A clinic specializes in testing for a disease D. The result of the test can be either positive or negative.

A study revealed that if a person suffers from the disease D, the test result in that clinic comes out positive 80% of the time, and negative 20% of the time. If a person is not suffering from the

disease D, the test comes out positive 10% of the time and negative 90% of the time. It is also known that among the general population, the disease D occurs in 30% of the individuals.

If a person tests positive for D in that clinic, the probability that he/she actually suffers from the disease D is _____. (Rounded off to two decimal places)

Q5 MSQ 2 Marks

Let X be a discrete valued random variable with cumulative distribution function $F(x)$.

Which of the following statements is/are correct?

( ) $F(x)$ is always a positive function.

( ) $F(x)$ is a non-decreasing function.

( ) $F(x)$ has jump discontinuity.

( ) $F(x)$ is a left continuous function.

Q6 MSQ 2 Marks

Let $X_1, X_2, \ldots, X_n$ be n independent random variables. Each of the random variables follows Normal($\mu = 0, \sigma^2 = 1$) distribution. Define $\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$.

Which of the following statements is/are correct?

( ) $\sum_{i=1}^{n} {X_i}^2$ follows Chi-square distribution with n degrees of freedom.

( ) $\sum_{i=1}^{n}(X_i - \bar{X})^2$ follows Chi-square distribution with $(n-1)$ degrees of freedom.

( ) ${X_1}^2 + {X_n}^2$ follows exponential distribution with mean 2.

( ) $(\sqrt{n}\bar{X})^2$ follows Chi-square distribution with 2 degrees of freedom.

Q7 MCQ 2 Marks

Let $L = \lim_{n\to\infty} \sum_{k=0}^{n} \frac{e^{-n}n^k}{k!}$

Which of the following is the value of L?

(A) 0.5

(B) 1.0

(C) 0

(D) e^{-1}

Q8 MCQ 2 Marks

Let X and Y be two independent random variables. X follows Bernoulli($p = 0.3$) distribution and Y follows Normal($\mu = 0, \sigma^2 = 100$) distribution.

Which of the following options is the variance of $(2X - 1)Y$?

(A) 100

(B) 90
(C) 49
(D) 21

Q9 NAT 1 Mark

Let X be an exponentially distributed random variable with mean $\lambda(>0)$. If $P(X > 5) = 0.35$, then the conditional probability $P(X > 10 \mid X > 5)$ is _____. (Rounded off to two decimal places)

Q10 MSQ 1 Mark

Suppose a random variable Z follows Normal$(\mu = 0, \sigma^2 = 1)$ distribution with probability density function $g(z)$ and cumulative distribution function $G(z)$. Another random variable Y follows t_1 distribution with probability density function $h(y)$ and cumulative distribution function $H(y)$. Let c be the positive real number for which $g(c) = h(c)$.

Which of the following statements is/are correct?

( ) $G(0) = H(0)$
( ) $G(c) < H(c)$
( ) $G(-c) < H(-c)$
( ) $g(0) = h(0)$

Q11 MCQ 1 Mark GATE DA 2026

Suppose that a computer program provides a non-negative and integer-valued random solution to the equation $n_1 + n_2 + n_3 + n_4 = 20$.

Which of the following is the probability that all of n_1, n_2, n_3, n_4 in the provided solution are positive?

(A) $\binom{19}{3} \Big/ \binom{23}{3}$
(B) $\binom{20}{4} \Big/ \binom{24}{4}$
(C) $\binom{20}{3} \Big/ \binom{23}{3}$
(D) $\binom{19}{4} \Big/ \binom{24}{4}$

Q12 MCQ 1 Mark GATE DA 2026

Let M be a randomly chosen non-empty subset of $S = \{1, 2, 3, \ldots, 2026\}$.

Which of the following is the probability that the product of all the elements of M is even?

(A) $2^{1013}(2^{1013} - 1)/2^{2026}$

(B) $2^{1013}/2^{2026}$
(C) $2^{1013}(2^{1013}-1)/(2^{2026}-1)$
(D) $1/(2^{2026}-1)$

Q13 NAT 2 Marks GATE DA 2025

A bag contains 5 white balls and 10 black balls. In a random experiment, n balls are drawn from the bag one at a time with replacement. Let S_n denote the total number of black balls drawn in the experiment. The expectation of S_{100} denoted by $E[S_{100}] =$ _____ (Round off to one decimal place)

Q14 MSQ 2 Marks GATE DA 2025

Consider a coin-toss experiment where the probability of head showing up is p. In the ith coin toss, let $X_i = 1$ if head appears, and $X_i = 0$ if tail appears. Consider

$$\hat{p} = \frac{1}{n}\sum_{i=1}^{n} X_i$$

where n is the total number of independent coin tosses. Which of the following statements is/are correct?

( ) $E[\hat{p}] = p$
( ) $E[\hat{p}] = \frac{p}{n}$
( ) As n increases, variance of $\hat{p}$ decreases
( ) Variance of $\hat{p}$ does not depend on n

Q15 MCQ 2 Marks GATE DA 2025

A random experiment consists of throwing 100 fair dice, each die having six faces numbered 1 to 6. An event A represents the set of all outcomes where at least one of the dice shows a 1. Then, $P(A) =$

(A) 0
(B) 1
(C) $1 - \left(\frac{5}{6}\right)^{100}$
(D) $\left(\frac{5}{6}\right)^{100}$

Q16 MCQ 2 Marks GATE DA 2025

For $x \in \mathbb{R}$, the floor function is denoted by $f(x) = \lfloor x \rfloor$ and defined as follows

$$\lfloor x \rfloor = k, \quad k \leq x < k+1,$$

where k is an integer. Let $Y = \lfloor X \rfloor$, where X is an exponentially distributed random variable with mean $\frac{1}{\ln 10}$, where ln denotes the natural logarithm. For any positive integer ℓ, one can write the probability of the event $Y = \ell$ as follows

$$P(Y = \ell) = q^{\ell}(1 - q)$$

The value of q is

(A) 0.1

(B) 0.01

(C) 0.5

(D) 0.434

Q17 MCQ 2 Marks GATE DA 2025

A random variable X is said to be distributed as Bernoulli(θ), denoted by $X \sim$ Bernoulli(θ), if

$$P(X = 1) = \theta, \quad P(X = 0) = 1 - \theta$$

for $0 < \theta < 1$. Let $Y = \sum_{i=1}^{300} X_i$, where $X_i \sim$ Bernoulli(θ), $i = 1, 2, \ldots, 300$ be independent and identically distributed random variables with $\theta = 0.25$. The value of $P(60 \leq Y \leq 90)$, after approximation through the Central Limit Theorem, is given by (Recall that $\phi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{x} e^{-\frac{t^2}{2}} dt$)

(A) $\phi(2) - \phi(-2)$

(B) $\phi(1) - \phi(-1)$

(C) $\phi(3) - \phi(-3)$

(D) $\phi(90) - \phi(60)$

Q18 MCQ 2 Marks GATE DA 2025

Consider the cumulative distribution function (CDF) of a random variable X:

$$F_X(x) = \begin{cases} 0, & x \leq -1 \\ \frac{1}{4}(x+1)^2, & -1 \leq x \leq 1 \\ 1, & x \geq 1 \end{cases}$$

The value of $P(X^2 \leq 0.25)$ is

(A) 0.625

(B) 0.25

(C) 0.5

(D) 0.5625

Q19 MCQ 2 Marks GATE DA 2025

Let $Y = Z^2$, $Z = \frac{X-\mu}{\sigma}$, where X is a normal random variable with mean μ and variance σ^2. The variance of Y is

(A) 1

(B) 2

(C) 3

(D) 4

Q20 NAT 1 Mark GATE DA 2025

There are three boxes containing white balls and black balls.

Box-1 contains 2 black and 1 white balls. Box-2 contains 1 black and 2 white balls. Box-3 contains 3 black and 3 white balls.

In a random experiment, one of these boxes is selected, where the probability of choosing Box-1 is $\frac{1}{2}$, Box-2 is $\frac{1}{6}$, and Box-3 is $\frac{1}{3}$. A ball is drawn at random from the selected box. Given that the ball drawn is white, the probability that it is drawn from Box-2 is _____ (Round off to two decimal places)

Q21 MCQ 1 Mark GATE DA 2025

It is given that $P(X \geq 2) = 0.25$ for an exponentially distributed random variable X with $E[X] = \frac{1}{\lambda}$, where $E[X]$ denotes the expectation of X. What is the value of λ? (ln denotes natural logarithm)

(A) $\ln 2$

(B) $\ln 4$

(C) $\ln 3$

(D) $\ln 0.25$

Q22 MCQ 1 Mark GATE DA 2025

Let $X = aZ + b$, where Z is a standard normal random variable, and a, b are two unknown constants. It is given that

$$E[X] = 1, \quad E[(X - E[X])Z] = -2, \quad E[(X - E[X])^2] = 4$$

where $E[X]$ denotes the expectation of random variable X. The values of a, b are:

(A) $a = -2, b = 1$

(B) $a = 2, b = -1$

(C) $a = -2, b = -1$

(D) $a = 1, b = 1$

Q23 MCQ 1 Mark GATE DA 2025

Let X be a continuous random variable whose cumulative distribution function (CDF) $F_X(x)$, for some t, is given as follows:

$$F_X(x) = \begin{cases} 0, & x \leq t \\ \frac{x-t}{4-t}, & t \leq x \leq 4 \\ 1, & x \geq 4 \end{cases}$$

If the median of X is 3, then what is the value of t?

(A) 2

(B) 1

(C) -1

(D) 0

Q24 MCQ 1 Mark GATE DA 2025

Suppose X and Y are random variables. The conditional expectation of X given Y is denoted by $E[X \mid Y]$. Then $E[E[X \mid Y]]$ equals

(A) $E[X \mid Y]$

(B) $\frac{E[X]}{E[Y]}$

(C) $E[X]$

(D) $E[Y]$

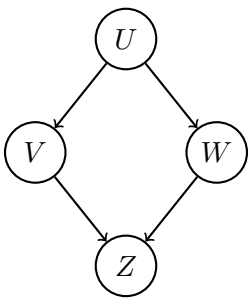
Q25 NAT 2 Marks GATE DA 2024

Two fair coins are tossed independently. X is a random variable that takes a value of 1 if both tosses are heads and 0 otherwise. Y is a random variable that takes a value of 1 if at least one of the tosses is heads and 0 otherwise.

The value of the covariance of X and Y is _____ (rounded off to three decimal places).

Q26 NAT 2 Marks GATE DA 2024

Given the following Bayesian Network consisting of four Bernoulli random variables and the associated conditional probability tables:



	$P(\cdot)$
$U = 0$	0.5
$U = 1$	0.5

	$P(V = 0 \mid \cdot)$	$P(V = 1 \mid \cdot)$
$U = 0$	0.5	0.5
$U = 1$	0.5	0.5

	$P(W = 0 \mid \cdot)$	$P(W = 1 \mid \cdot)$
$U = 0$	1	0
$U = 1$	0	1

		$P(Z = 0 \mid \cdot)$	$P(Z = 1 \mid \cdot)$
$V = 0$	$W = 0$	0.5	0.5
$V = 0$	$W = 1$	1	0
$V = 1$	$W = 0$	1	0
$V = 1$	$W = 1$	0.5	0.5

The value of $P(U = 1, V = 1, W = 1, Z = 1) =$ _____ (rounded off to three decimal places).

Q27 NAT 2 Marks GATE DA 2024

Consider a joint probability density function of two random variables X and Y

$$f_{X,Y}(x, y) = \begin{cases} 2xy, & 0 < x < 2, \quad 0 < y < x \\ 0, & \text{otherwise} \end{cases}$$

Then, $E[Y \mid X = 1.5]$ is _____.

Q28 NAT 2 Marks GATE DA 2024

Consider two events T and S. Let $\bar{T}$ denote the complement of the event T. The probability associated with different events are given as follows:

$$P(\bar{T}) = 0.6, \quad P(S \mid T) = 0.3, \quad P(S \mid \bar{T}) = 0.6$$

Then, $P(T \mid S)$ is _____ (rounded off to two decimal places).

Q29 NAT 2 Marks GATE DA 2024

Let X be a random variable exponentially distributed with parameter $\lambda > 0$. The probability density function of X is given by:

$$f_X(x) = \begin{cases} \lambda e^{-\lambda x}, & x \geq 0 \\ 0, & \text{otherwise} \end{cases}$$

If $5E(X) = \text{Var}(X)$, where $E(X)$ and $\text{Var}(X)$ indicate the expectation and variance of X, respectively, the value of λ is _____ (rounded off to one decimal place).

Q30 NAT 2 Marks GATE DA 2024

Let X be a random variable uniformly distributed in the interval $[1, 3]$ and Y be a random variable uniformly distributed in the interval $[2, 4]$. If X and Y are independent of each other, the probability $P(X \geq Y)$ is _____ (rounded off to three decimal places).

Q31 MCQ 2 Marks GATE DA 2024

A fair six-sided die (with faces numbered $1, 2, 3, 4, 5, 6$) is repeatedly thrown independently. What is the expected number of times the die is thrown until two consecutive throws of even numbers are seen?

(A) 2
(B) 4
(C) 6
(D) 8

Q32 NAT 1 Mark GATE DA 2024

The sample average of 50 data points is 40. The updated sample average after including a new data point taking the value of 142 is _____.

Q33 MCQ 1 Mark GATE DA 2024

Let the minimum, maximum, mean and standard deviation values for the attribute income of data scientists be Rs. 46000, Rs. 170000, Rs. 96000, and Rs. 21000, respectively. The z-score normalized income value of Rs. 106000 is closest to which ONE of the following options?

(A) 0.217
(B) 0.476
(C) 0.623
(D) 2.304

Q34 MCQ 1 Mark GATE DA 2024

Consider five random variables U, V, W, X, and Y whose joint distribution satisfies:

$$P(U, V, W, X, Y) = P(U)P(V)P(W \mid U, V)P(X \mid W)P(Y \mid W)$$

Which ONE of the following statements is FALSE?

(A) Y is conditionally independent of V given W
(B) X is conditionally independent of U given W
(C) U and V are conditionally independent given W
(D) Y and X are conditionally independent given W

Q35 MCQ 1 Mark GATE DA 2024

Three fair coins are tossed independently. T is the event that two or more tosses result in heads. S is the event that two or more tosses result in tails.

What is the probability of the event $T \cap S$?

(A) 0

(B) 0.5

(C) 0.25

(D) 1

Q36 MCQ 1 Mark GATE DA 2024

Consider the following statements:

(i) The mean and variance of a Poisson random variable are equal. (ii) For a standard normal random variable, the mean is zero and the variance is one.

Which ONE of the following options is correct?

(A) Both (i) and (ii) are true

(B) (i) is true and (ii) is false

(C) (ii) is true and (i) is false

(D) Both (i) and (ii) are false

Worked Solutions

GATE DA Probability and Statistics

Step-by-step methods, key formulas, and answer checks

Quick Answer Key

Q	Answer	Q	Answer
1	0.10	19	(B) 2
2	0	20	0.25
3	1	21	(A) $\ln 2$
4	0.77	22	(A) $a = -2,\ b = 1$
5	(B), (C)	23	(A) 2
6	(A), (B), (C)	24	(C) $E[X]$
7	(A) 0.5	25	0.063
8	(A) 100	26	0.125
9	0.35	27	1.0 (conditional calculation)
10	(A), (C)	28	0.25
11	(A)	29	0.2
12	(C)	30	0.125
13	66.7	31	(C) 6
14	(A), (C)	32	42
15	(C)	33	(B) 0.476
16	(A) 0.1	34	(C) is false
17	(A)	35	(A) 0
18	(C) 0.5	36	(A)

Note: The solutions below follow the questions as supplied. Q27's stated joint density does not integrate to 1 over its support, so it is not a valid joint PDF as written. Its conditional-density ratio nevertheless gives the requested conditional expectation.

Detailed Solutions

Q1 NAT 2 Marks Bernoulli matrix

Let $X = A_{2,3}$ be the shared entry in row 2 and column 3. If $X = 1$, the other four entries in each of the row and column must contain exactly two ones; this has probability $\binom{4}{2}/2^4 = 6/16$ for each, independently. If $X = 0$, each set of four entries must contain exactly three ones; probability

$\binom{4}{3}/2^4 = 4/16$ for each. Therefore,

$$P = \tfrac{1}{2}\left(\tfrac{6}{16}\right)^2 + \tfrac{1}{2}\left(\tfrac{4}{16}\right)^2 = \frac{13}{128} \approx 0.10156.$$

Rounded to two decimal places: $\boxed{0.10}$.

Q2 NAT 2 Marks Correlation

By the conditional-uniform model, $E[Y \mid X] = X^2$. Hence $E[Y] = E[X^2] = 1/3$ and

$$E[XY] = E\{XE[Y \mid X]\} = E[X^3] = 0,$$

by symmetry of $X \sim U(-1, 1)$. Thus $\mathrm{Cov}(X, Y) = E[XY] - E[X]E[Y] = 0$. Since both variables have nonzero variance, $\boxed{\mathrm{Corr}(X, Y) = 0}$.

Q3 NAT 2 Marks Pairwise differences

Use the identity

$$\sum_{i=1}^{n}\sum_{j=1}^{n}(x_i - x_j)^2 = 2n\sum_{i=1}^{n}(x_i - \bar{x})^2.$$

With $n = 100$, the left side divided by $2000 = 2n$ is exactly $\sum_i (x_i - \bar{x})^2 = 99$. Therefore

$$\frac{1}{99}\sum_{i=1}^{100}(x_i - \bar{x})^2 = \boxed{1}.$$

Q4 NAT 2 Marks Bayes' theorem

Let D denote disease and $+$ a positive result. Bayes' theorem gives

$$P(D \mid +) = \frac{P(+ \mid D)P(D)}{P(+ \mid D)P(D) + P(+ \mid D^c)P(D^c)} = \frac{0.8(0.3)}{0.8(0.3) + 0.1(0.7)} = \frac{0.24}{0.31} \approx 0.7742.$$

Rounded to two decimal places: $\boxed{0.77}$.

Q5 MSQ 2 Marks CDF properties

A CDF satisfies $0 \le F(x) \le 1$, is non-decreasing, and is right-continuous. For a discrete random variable it has jumps at points with positive probability. Thus "non-decreasing" and "has jump discontinuity" are correct: $\boxed{\text{(B), (C)}}$. "Always positive" is false because a CDF can equal zero, and a CDF is right-continuous, not generally left-continuous.

Q6 MSQ 2 Marks Normal and chi-square

For independent standard normal variables, $\sum_{i=1}^{n} X_i^2 \sim \chi_n^2$, so (A) is true. Also, $\sum_i (X_i - \bar{X})^2 \sim \chi_{n-1}^2$, so (B) is true. Since $X_1^2 + X_n^2 \sim \chi_2^2$, it is exponential with mean 2, so (C) is true (assuming the two indexed variables are distinct independent observations). Finally, $\sqrt{n}\bar{X} \sim N(0,1)$, so its square is χ_1^2, not χ_2^2. Answer: $\boxed{\text{(A), (B), (C)}}$.

Q7 MCQ 2 Marks Poisson limit

The sum is the cumulative probability $P(N \le n)$ for $N \sim \text{Poisson}(n)$. A Poisson variable with mean n has standard deviation $\sqrt{n}$; by the central limit theorem, the probability of being at or below its mean tends to 1/2. Hence $\boxed{\text{(A) 0.5}}$.

Q8 MCQ 2 Marks Variance of a product

Since X is Bernoulli, $(2X-1)$ is either -1 or 1, so $(2X-1)^2 = 1$. Independence and $E[Y] = 0$ give

$$\text{Var}((2X-1)Y) = E[(2X-1)^2 Y^2] = E[Y^2] = \text{Var}(Y) = 100.$$

Answer: $\boxed{\text{(A) 100}}$.

Q9 NAT 1 Mark Memoryless property

An exponential random variable is memoryless:

$$P(X > 10 \mid X > 5) = P(X > 10 - 5) = P(X > 5) = \boxed{0.35}.$$

Q10 MSQ 1 Mark Normal versus Student t_1

Both distributions are symmetric about zero, so $G(0) = H(0) = 1/2$ and (A) is true. The standard normal has lighter tails than the Cauchy (t_1), so for $c > 0$ the normal CDF is larger: $G(c) > H(c)$, making (B) false. By symmetry, $G(-c) = 1 - G(c) < 1 - H(c) = H(-c)$, so (C) is true. The densities at zero are $g(0) = 1/\sqrt{2\pi}$ and $h(0) = 1/\pi$, which are unequal. Answer: $\boxed{\text{(A), (C)}}$.

Q11 MCQ 1 Mark Stars and bars

The number of non-negative integer solutions of $n_1 + n_2 + n_3 + n_4 = 20$ is $\binom{20+4-1}{4-1} = \binom{23}{3}$. For all four variables to be positive, set $m_i = n_i - 1$; then $m_1 + \cdots + m_4 = 16$, giving $\binom{19}{3}$ solutions. The required probability is $\boxed{\binom{19}{3} / \binom{23}{3}}$, option (A).

Q12 MCQ 1 Mark Random subsets

There are $2^{2026} - 1$ non-empty subsets. The product is odd exactly when every selected element is odd. There are 1013 odd elements, so there are $2^{1013} - 1$ non-empty odd-only subsets. Thus

$$P(\text{even product}) = 1 - \frac{2^{1013} - 1}{2^{2026} - 1} = \frac{2^{2026} - 2^{1013}}{2^{2026} - 1} = \boxed{\text{(C)}}.$$

Q13 NAT 2 Marks Expectation of a binomial count

Each draw is black with probability $10/(5 + 10) = 2/3$. With replacement, $S_{100} \sim \text{Binomial}(100, 2/3)$, so $E[S_{100}] = np = 100(2/3) = 66.666\ldots$. Rounded to one decimal place: $\boxed{66.7}$.

Q14 MSQ 2 Marks Sample proportion

For independent Bernoulli(p) variables, $E[X_i] = p$ and $\text{Var}(X_i) = p(1 - p)$. Therefore

$$E[\hat{p}] = \frac{1}{n}\sum_i E[X_i] = p, \qquad \text{Var}(\hat{p}) = \frac{p(1-p)}{n}.$$

The expectation is unbiased and the variance decreases as n increases. Answer: $\boxed{\text{(A), (C)}}$.

Q15 MCQ 2 Marks Complement rule

It is easier to compute the complement: no die shows a 1 with probability $(5/6)^{100}$. Hence

$$P(\text{at least one 1}) = 1 - \left(\frac{5}{6}\right)^{100}.$$

Answer: $\boxed{\text{(C)}}$.

Q16 MCQ 2 Marks Floor of an exponential variable

The exponential mean is $1/\lambda = 1/\ln 10$, so $\lambda = \ln 10$. For integer $\ell \geq 0$,

$$P(\lfloor X \rfloor = \ell) = P(\ell \leq X < \ell + 1) = e^{-\lambda\ell}(1 - e^{-\lambda}) = (0.1)^{\ell}(0.9).$$

Comparing with $q^{\ell}(1 - q)$ gives $\boxed{q = 0.1}$, option (A).

Q17 MCQ 2 Marks Central limit theorem

For $Y \sim \text{Binomial}(300, 0.25)$, $\mu = np = 75$ and $\sigma = \sqrt{np(1-p)} = \sqrt{56.25} = 7.5$. The bounds 60 and 90 are 2 standard deviations below and above the mean. Thus the CLT approximation is

$\Phi(2) - \Phi(-2)$, option $\boxed{\text{(A)}}$.

Q18 MCQ 2 Marks Probability from a CDF

The event $X^2 \le 0.25$ is $-0.5 \le X \le 0.5$. Using the supplied CDF,

$$P(-0.5 \le X \le 0.5) = F(0.5) - F(-0.5) = \frac{(1.5)^2}{4} - \frac{(0.5)^2}{4} = 0.5625 - 0.0625 = 0.5.$$

Answer: $\boxed{\text{(C) } 0.5}$.

Q19 MCQ 2 Marks Variance of a squared normal

The standardized variable $Z = (X - \mu)/\sigma$ is standard normal, and $Y = Z^2 \sim \chi^2_1$. A chi-square variable with k degrees of freedom has variance $2k$, so $\mathrm{Var}(Y) = 2$. Answer: $\boxed{\text{(B)}}$.

Q20 NAT 1 Mark Bayes' theorem with boxes

The joint probabilities of choosing a box and drawing white are

$$P(B_1, W) = \tfrac{1}{2} \cdot \tfrac{1}{3} = \tfrac{1}{6}, \quad P(B_2, W) = \tfrac{1}{6} \cdot \tfrac{2}{3} = \tfrac{1}{9}, \quad P(B_3, W) = \tfrac{1}{3} \cdot \tfrac{1}{2} = \tfrac{1}{6}.$$

Thus $P(W) = 1/6 + 1/9 + 1/6 = 4/9$ and

$$P(B_2 \mid W) = \frac{1/9}{4/9} = \boxed{0.25}.$$

Q21 MCQ 1 Mark Exponential rate

For an exponential variable with rate λ, $P(X \ge x) = e^{-\lambda x}$. Hence $e^{-2\lambda} = 0.25 = 1/4$, so $2\lambda = \ln 4$ and $\lambda = \ln 2$. Answer: $\boxed{\text{(A)}}$.

Q22 MCQ 1 Mark Affine transformation of a normal

Since $X = aZ + b$ and $E[Z] = 0$, $E[X] = b = 1$. Also, $E[(X - E[X])Z] = E[aZ^2] = a = -2$. The variance is $a^2 = 4$, consistent with the supplied value. Therefore $\boxed{a = -2,\ b = 1}$, option (A).

Q23 MCQ 1 Mark Median of a uniform variable

The given CDF describes a uniform distribution on $[t, 4]$. Its median is the midpoint $(t + 4)/2$. Setting this equal to 3 gives $t + 4 = 6$, so $\boxed{t = 2}$, option (A).

Q24 MCQ 1 Mark Law of total expectation

The tower property states $E[E[X \mid Y]] = E[X]$, provided the expectation exists. Answer: $\boxed{\text{(C)}}$.

Q25 NAT 2 Marks Covariance of indicators

$X = 1$ only for HH, while $Y = 1$ for HH, HT, and TH. Since $X = 1$ implies $Y = 1$, $XY = X$. Therefore $E[XY] = 1/4$, $E[X] = 1/4$, and $E[Y] = 3/4$. Hence

$$\mathrm{Cov}(X, Y) = E[XY] - E[X]E[Y] = \frac{1}{4} - \frac{1}{4} \cdot \frac{3}{4} = \frac{1}{16} = 0.0625.$$

Rounded to three decimal places: $\boxed{0.063}$.

Q26 NAT 2 Marks Bayesian network

Factorize using the network structure:

$$P(U, V, W, Z) = P(U)P(V \mid U)P(W \mid U)P(Z \mid V, W).$$

From the tables, $P(U = 1) = 0.5$, $P(V = 1 \mid U = 1) = 0.5$, $P(W = 1 \mid U = 1) = 1$, and $P(Z = 1 \mid V = 1, W = 1) = 0.5$. Therefore

$$P(1, 1, 1, 1) = 0.5 \times 0.5 \times 1 \times 0.5 = \boxed{0.125}.$$

Q27 NAT 2 Marks Conditional expectation

For a fixed $x > 0$, integrate the supplied expression over $0 < y < x$:

$$f_X(x) = \int_0^x 2xy \, dy = x^3.$$

The conditional-density ratio is therefore $f_{Y|X}(y \mid x) = 2y/x^2$ for $0 < y < x$. Consequently,

$$E[Y \mid X = x] = \int_0^x y \frac{2y}{x^2} \, dy = \frac{2x}{3}.$$

At $x = 1.5$, this is $\boxed{1.0}$. Important source check: the stated joint function integrates to 4, not 1, over $0 < x < 2$, $0 < y < x$; it is not a valid joint PDF as written. The conditional ratio above still yields the indicated answer if the intended expression is used formally.

Q28 NAT 2 Marks Conditional probability

Given $P(\bar{T}) = 0.6$, we have $P(T) = 0.4$. By total probability,

$$P(S) = P(S \mid T)P(T) + P(S \mid \bar{T})P(\bar{T}) = 0.3(0.4) + 0.6(0.6) = 0.48.$$

Thus $P(T \mid S) = P(S \mid T)P(T)/P(S) = 0.12/0.48 = \boxed{0.25}$.

Q29 NAT 2 Marks Mean and variance of exponential

For an exponential variable with rate λ, $E[X] = 1/\lambda$ and $\mathrm{Var}(X) = 1/\lambda^2$. The condition $5E[X] = \mathrm{Var}(X)$ gives $5/\lambda = 1/\lambda^2$, hence $\lambda = 1/5 = \boxed{0.2}$.

Q30 NAT 2 Marks Probability by area

The rectangle supporting (X, Y) has area $(3-1)(4-2) = 4$. The event $X \geq Y$ is possible only for $2 \leq x \leq 3$ and $2 \leq y \leq x$. Its area is

$$\int_2^3 (x-2)\,dx = \frac{1}{2}.$$

Therefore $P(X \geq Y) = (1/2)/4 = \boxed{0.125}$.

Q31 MCQ 2 Marks Expected waiting time

Let E_0 be the expected number of further throws when the previous throw is not an even number (or at the start), and E_1 when the previous throw was even. Each throw is even with probability $1/2$. Then

$$E_0 = 1 + \tfrac{1}{2}E_1 + \tfrac{1}{2}E_0, \qquad E_1 = 1 + \tfrac{1}{2}E_0,$$

where an even throw from state E_1 completes the target and needs no further throws. The first equation gives $E_0 = 2 + E_1$. Substituting yields $E_0 = 2 + 1 + E_0/2$, so $E_0 = 6$. Answer: $\boxed{\text{(C) } 6}$.

Q32 NAT 1 Mark Updated mean

The original sum is $50 \times 40 = 2000$. Including the new value gives total 2142 across 51 points, so the updated mean is $2142/51 = \boxed{42}$.

Q33 MCQ 1 Mark Z-score normalization

The z-score is $z = (x - \mu)/\sigma$. Thus

$$z = \frac{106000 - 96000}{21000} = \frac{10000}{21000} \approx 0.4762.$$

The closest choice is $\boxed{\text{(B) } 0.476}$.

Q34 MCQ 1 Mark Conditional independence

The factorization is $P(U)P(V)P(W \mid U,V)P(X \mid W)P(Y \mid W)$. It implies $X \perp U \mid W$, $Y \perp V \mid W$, and $X \perp Y \mid W$. However, U and V need not be conditionally independent given their common child W; conditioning on a common effect can induce dependence. Therefore the false statement is $\boxed{\text{(C)}}$.

Q35 MCQ 1 Mark Events on three coin tosses

For three tosses, "at least two heads" and "at least two tails" cannot both occur: the total number of heads and tails is three, so they cannot each be at least two. Therefore $T \cap S = \varnothing$ and $P(T \cap S) = \boxed{0}$, option (A).

Q36 MCQ 1 Mark Basic distribution facts

For $X \sim \text{Poisson}(\lambda)$, $E[X] = \text{Var}(X) = \lambda$. A standard normal variable has mean 0 and variance 1. Both statements are true, so the answer is $\boxed{\text{(A)}}$.

Important Formula Sheet

Expectation and variance

$$E[aX + b] = aE[X] + b, \quad \mathrm{Var}(aX + b) = a^2 \,\mathrm{Var}(X),$$

$$\mathrm{Cov}(X, Y) = E[XY] - E[X]E[Y], \quad \rho_{XY} = \frac{\mathrm{Cov}(X, Y)}{\sigma_X \sigma_Y}.$$

Conditional expectation and Bayes' theorem

$$E[E[X \mid Y]] = E[X], \quad P(A \mid B) = \frac{P(B \mid A)P(A)}{P(B)}.$$

Common distributions

Distribution	$E[X]$	$\mathrm{Var}(X)$
Bernoulli(p)	p	$p(1-p)$
Binomial(n, p)	np	$np(1-p)$
Poisson(λ)	λ	λ
Exponential(λ rate)	$1/\lambda$	$1/\lambda^2$
$N(\mu, \sigma^2)$	μ	σ^2

Useful identities

$$\mathrm{Var}(X) = E[X^2] - (E[X])^2, \quad \sum_{i,j}(x_i - x_j)^2 = 2n \sum_i (x_i - \bar{x})^2.$$

For $N \sim \mathrm{Poisson}(\lambda)$, N is approximately normal with mean λ and variance λ for large λ. For $Y \sim \mathrm{Binomial}(n, p)$, the CLT approximation uses mean np and standard deviation $\sqrt{np(1-p)}$.

End of Question Set and Worked Solutions